

innovative · active · music · experienced
Researcher · Listener · Composer · Collaborator



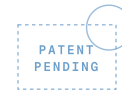
Connecting technology with the people it serves (B.A. Digital Culture).

sEEG Sonification Toolkit v8.3 – translating intracranial EEG into navigable sound. *“Impressive and very interesting.”* – Teunis van Beelen, creator of EDFbrowser.



Inventing a framework for human tempo (provisional patent filed).

Relative Tempo Derivation (RTD) – biomarker-to-tempo mapping rooted in just-intonation ratios.

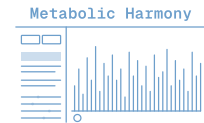


相對節奏推導



Researcher and composer exploring new ways of using harmony.

Metabolic Harmony – an interactive web sonification that lets you hear the gut microbiome. 350 metabolic pathways rendered as harmonic ratios across 4,869 samples, with built-in comparison between healthy reference and Multiple Sclerosis states – built to make complex pathway data audible, and accessible to researchers and patients alike.



[EEG & Auditory Neuroscience Work Summary →](#)

RESEARCH, COMPOSITION & COACHING

Independent Researcher & Composer

Lifelong Practice

Self-directed research and creative practice. Original frameworks in data sonification methodology, harmonic theory, and audio tools. Collaborates across disciplines, managing creative input from multiple contributors to produce cohesive final work. Independently releases musical projects.

Percussion Coach, Arranger & Educator

Casa Grande HS · Villago MS · Flux Indoor · 2013-2022

Six Arizona state championships with Casa Grande HS (2015-2020). Wrote and arranged original competitive percussion scores; integrated live sound – microphones, synths, samplers, mixing – for full production value. Private percussion instruction. Performer and section contributor with ASU World Drumline and The Academy Drum & Bugle Corps.

EDUCATION & TRAINING

Bachelor of Arts, Digital Culture

Arizona State University · 2017

Arts, Media & Engineering. Technology integration, digital systems, human-computer interaction. Direct academic foundation for audio technology, signal processing, and interactive media research.

Neurologic Music Therapy – Volunteer Observer

NMTSA · 2025 – Present

Observed clinical NMT sessions with detailed protocol notes. Direct exposure to therapeutic audio applications including rhythmic auditory stimulation in stroke gait rehabilitation.

ADDITIONAL EXPERIENCE

Market Hand & Sales Partner

Bene Vivendo · 2023 – Present

Small-team operation working directly with the owner. Customer education, product presentation, sales, and community engagement at weekly farmers market. Hands-on experience in the rhythms of a founder-run business.

Aircraft Security Check

Phoenix Sky Harbor · Current

Situational awareness and effective work under strenuous, time-sensitive conditions.

Administrative Assistant

ASU Engineering Technical Services · 2013-2017

Front desk and administrative support for the office that supports research operations for ASU engineering faculty and staff.

Barback

Local bar · Phoenix, AZ

Fast-paced service environment. Team coordination, stock management, and customer-facing work in a high-volume setting.

REFERENCES

Leanna Moore

Education Director, NMTSA
Contact available on request

Emily Heller

Owner, Bene Vivendo
Contact available on request

Matt Geres

Coordinator, Casa Grande HS
Contact available on request